HW5

Monday, 24 March 2025 2:34 PM

Problem 1

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} = \frac{\partial^2 u}{\partial y^2}$$

$(x, y, t) \rightarrow (\xi, \eta, \tau)$

(a) We can use coordinate transformations: (using $t = \tau$ & $x \neq f(x,y)$)

$$\frac{\partial}{\partial x} = \xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial y} = \xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial \tau} = \frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}$$

term 1 term 2 term 3 term 4

$$\therefore \boxed{\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right) u + \left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right) u^2 + \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right) uv = \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)^2 u}$$

Chain-Rule Conservation-Law form

(b) To maintain conservation have to convert to Strong Conservation-Law form

1) $J \times$ (PDE from above)

2) Move J term inside via chain rule

Term 1:

$$\therefore J\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right) u = J\frac{\partial u}{\partial \tau} + J\xi_t \frac{\partial u}{\partial \xi} + J\eta_t \frac{\partial u}{\partial \eta}$$

$$= \left(\frac{\partial (Ju)}{\partial \tau} - u\frac{\partial J}{\partial \tau}\right) + \frac{\partial (J\xi_t u)}{\partial \xi} - u\frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - u\frac{\partial (J\eta_t)}{\partial \eta}$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \left(u\frac{\partial J}{\partial \tau} + u\frac{\partial (J\xi_t)}{\partial \xi} + u\frac{\partial (J\eta_t)}{\partial \eta}\right)$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \underbrace{\left(\frac{\partial J}{\partial \tau} + \frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t)}{\partial \eta}\right)}_{0 \text{ (geometric conservation law)}} u$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi}^{0} + \frac{\partial (J\eta_t u)}{\partial \eta}^{0} = \frac{\partial (Ju)}{\partial \tau}$$

Similar manipulation of term 2, & term 3 (first order spatial derivatives)

term 2:

$$J\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right) u^2 = \frac{\partial (J\xi_x u^2)}{\partial \xi} + \frac{\partial (J\eta_x u^2)}{\partial \eta} - u^2 \underbrace{\left(\frac{\partial (J\xi_x)}{\partial \xi} + \frac{\partial (J\eta_x)}{\partial \eta}\right)}_{=0 \text{ (geometric conservation law)}}$$

term 3:

$$J\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right) uv = \frac{\partial (J\xi_y uv)}{\partial \xi} + \frac{\partial (J\eta_y uv)}{\partial \eta} - uv \underbrace{\left(\frac{\partial (J\xi_y)}{\partial \xi} + \frac{\partial (J\eta_y)}{\partial \eta}\right)}_{=0 \text{ (geometric cons. law)}}$$

term 4:

$$J\frac{1}{J}\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)^2 u = \frac{1}{J}\left(J\xi_y \frac{\partial}{\partial \xi} + J\eta_y \frac{\partial}{\partial \eta}\right)^2 u$$

$$= \frac{1}{J}\left(J^2\xi_y^2 \frac{\partial^2}{\partial \xi^2} + J^2\eta_y^2 \frac{\partial^2}{\partial \eta^2} + 2J^2\xi_y\eta_y \frac{\partial^2}{\partial \xi \partial \eta}\right) u$$

$$= \frac{1}{J}\left(\frac{\partial}{\partial \xi}(J^2\xi_y^2 u) - u\frac{\partial}{\partial \xi}(J^2\xi_y^2) + \frac{\partial}{\partial \eta}(J^2\eta_y^2 u) - u\frac{\partial}{\partial \eta}(J^2\eta_y^2) + 2\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y u) - 2u\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y)\right)$$

$$= \frac{1}{J}\left[\frac{\partial^2}{\partial \xi^2}(J^2\xi_y^2 u) + \frac{\partial^2}{\partial \eta^2}(J^2\eta_y^2 u) + 2\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y u)\right] - \frac{u}{J}\underbrace{\left(\frac{\partial^2}{\partial \xi^2}(J^2\xi_y^2) + \frac{\partial^2}{\partial \eta^2}(J^2\eta_y^2) + 2\frac{\partial^2}{\partial \xi \partial \eta}(J^2\xi_y\eta_y)\right)}_{0 \text{ (= geometric conservation law)}}$$

Therefore,

$J\left[\frac{\partial \xi}{\partial x} \quad \frac{\partial \xi}{\partial y} \quad \frac{\partial \eta}{\partial x} \quad \frac{\partial \eta}{\partial y}\right]$ $J\left[\frac{\partial}{\partial \xi}\left(\frac{\partial x}{\partial \eta}\right) \quad \frac{\partial}{\partial \eta}\left(\frac{\partial x}{\partial \eta}\right) \quad \frac{\partial}{\partial \xi}\left(\frac{\partial x}{\partial \xi}\right)\right]$

0 (= geometric conservation law)

Therefore,

Final strong conservative form:

$$\frac{\partial (Ju)}{\partial \tau} + \left[\frac{\partial}{\partial \xi}(J\xi_x u) + \frac{\partial}{\partial \eta}(J\eta_x u)\right] + \left[\frac{\partial}{\partial \xi}(J\xi_y u) + \frac{\partial}{\partial \eta}(J\eta_y u)\right] = \frac{\partial^2}{\partial \xi^2}(J^2 \xi_y^2 u) + \frac{\partial^2}{\partial \eta^2}(J^2 \eta_y^2 u) + \frac{\partial^2}{\partial \xi \partial \eta}(J^2 \xi_y \eta_y u)$$

$$\boxed{\frac{\partial (Ju)}{\partial \tau} + \frac{\partial}{\partial \xi}(J\xi_x u^2 + J\xi_y uv) + \frac{\partial}{\partial \eta}(J\eta_x u^2 + J\eta_y uv) = \frac{\partial^2}{\partial \xi^2}(J^2 \xi_y^2 u) + \frac{\partial^2}{\partial \eta^2}(J^2 \eta_y^2 u) + \frac{\partial^2}{\partial \xi \partial \eta}(J^2 \xi_y \eta_y u)}$$

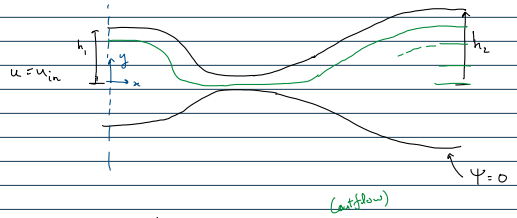
(c) Highlighted terms in blue show the 4 geometric conservation laws (in expression above)

Problem 2

Incompressible potential flow $\nabla^2 \psi = 0$ (ψ: stream function)

(a) Describe the problem

(b) Governing equations



$u = \frac{\partial \psi}{\partial y}$ $v = -\frac{\partial \psi}{\partial x}$

2D, axisymmetric flow ; ψ = stream function (constant along a stream line)

$\therefore \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$

governing equations

$\frac{\partial}{\partial x}(-v) + \frac{\partial u}{\partial y} = 0$

$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0$

BC's:

(Inflow) $u_{in} = \left.\frac{\partial \psi}{\partial y}\right|_{x=0}$

(outflow) $\left.\frac{\partial \psi}{\partial x}\right|_{x=c} = 0$

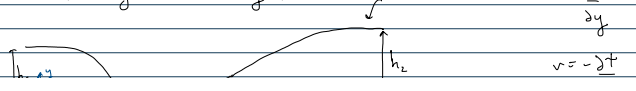
(walls) $\psi = 0$ (@ curve 1)

$\psi = u_{in} h_1$ (@ curve 2)

$$\int_{(y-h_1)}^{(y+h_1)} u_{in}\, dy = \int_{\psi=0}^{\psi(y)} d\psi$$

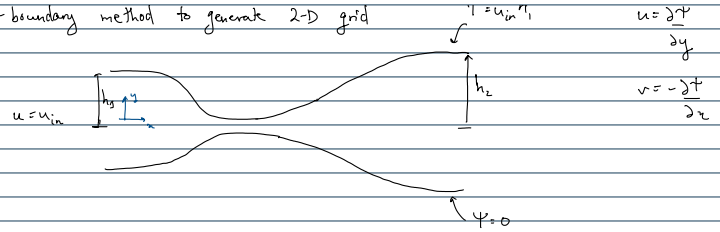
$$\psi(y) = u_{in}\left((y+h_1) - (y-h_1)\right) = u_{in}(2h_1)$$

(c) Two-boundary method to generate 2-D grid



$u = \frac{\partial \psi}{\partial y}$ $v = -\frac{\partial \psi}{\partial x}$

(c) Two-boundary method to generate 2-D grid



$$u = \frac{\partial \psi}{\partial y} \qquad v = -\frac{\partial \psi}{\partial x}$$

$$u_{in} = \left.\frac{\partial \psi}{\partial y}\right|_{x=0}$$

$$\int_{-h_1}^{h_1} u_{in}\, dy = \int_{\psi=0}^{\psi(y)} d\psi$$

(d) Transform (x, y) to (ξ, η)

$$\nabla^2 \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$J^2 \left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta} \right)^2 \psi + J^2 \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta} \right)^2 \psi = 0$$

Strong Conservative

$$\Rightarrow \left[\frac{\partial^2 (J^2 \xi_x^2 \psi)}{\partial \xi^2} + \frac{\partial^2 (J^2 \eta_x^2 \psi)}{\partial \eta^2} + 2\frac{\partial}{\partial \xi}\left(J^2 \xi_x \eta_x^2 \psi\right) \right] + \left[\frac{\partial^2 (J^2 \xi_y^2 \psi)}{\partial \xi^2} + \frac{\partial^2 (J^2 \eta_y^2 \psi)}{\partial \eta^2} + 2\frac{\partial}{\partial \xi}\left(J^2 \xi_y \eta_y^2 \psi\right) \right]$$

$$- \psi \left[\frac{\partial^2 (J^2 \xi_x^2)}{\partial \xi^2} + \frac{\partial^2 (J^2 \eta_x^2)}{\partial \eta^2} + 2\frac{\partial}{\partial \xi}\left(J^2 \xi_x \eta_x^2\right) + 2\frac{\partial}{\partial \eta}\left(J^2 \xi_y^2 \eta_y^2\right) \right] = 0 \ ?$$

$$\Rightarrow \frac{\partial^2}{\partial \xi^2}\left(J^2 \xi_x^2 \psi + J^2 \xi_y^2 \psi\right) + \frac{\partial^2}{\partial \eta^2}\left(J^2 \eta_x^2 \psi + J^2 \eta_y^2 \psi\right) + 2\frac{\partial}{\partial \xi \partial \eta}\left(J^2 \xi_x \eta_x^2 + J^2 \xi_y^2 \eta_y^2\right) = 0$$

y_n^2 $\qquad \psi_{i+1} - 2\psi + \psi_{-1}$

$y_n = \xi_x \cdot J$

(e), (f) FDE/FVE's for governing equations + algorithm

$$\frac{\partial^2 \psi}{\partial \xi^2} = \frac{\psi_{i+1,j} - 2\psi_{ij} + \psi_{i-1,j}}{\Delta \xi^2}$$

$$\frac{\partial^2 \psi}{\partial \eta^2} = \frac{\psi_{i,j+1} - 2\psi_{ij} + \psi_{i,j-1}}{\Delta \eta^2}$$

$$\xi_x = \frac{\partial \xi}{\partial x} = \frac{y_\eta}{J}$$

$$\frac{\partial^2 \psi}{\partial \xi \partial \eta} = \frac{\psi_{i+1,j+1} - \psi_{i+1,j-1} - \psi_{i-1,j+1} + \psi_{i-1,j-1}}{4 \Delta \xi \Delta \eta}$$

(g), (h) Code + solutions for velocity = 100 m/s

(i) Discuss accuracy of results.